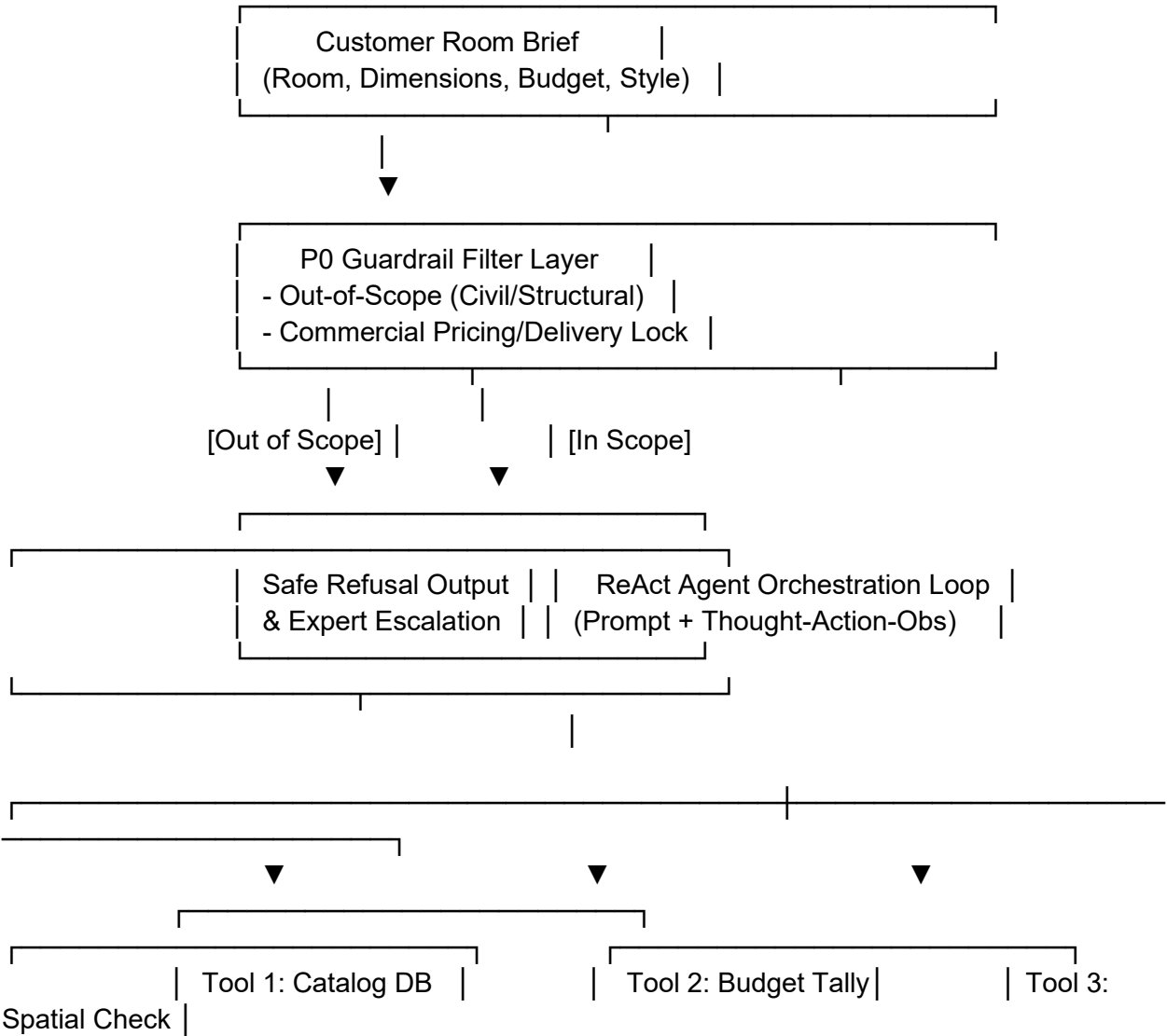
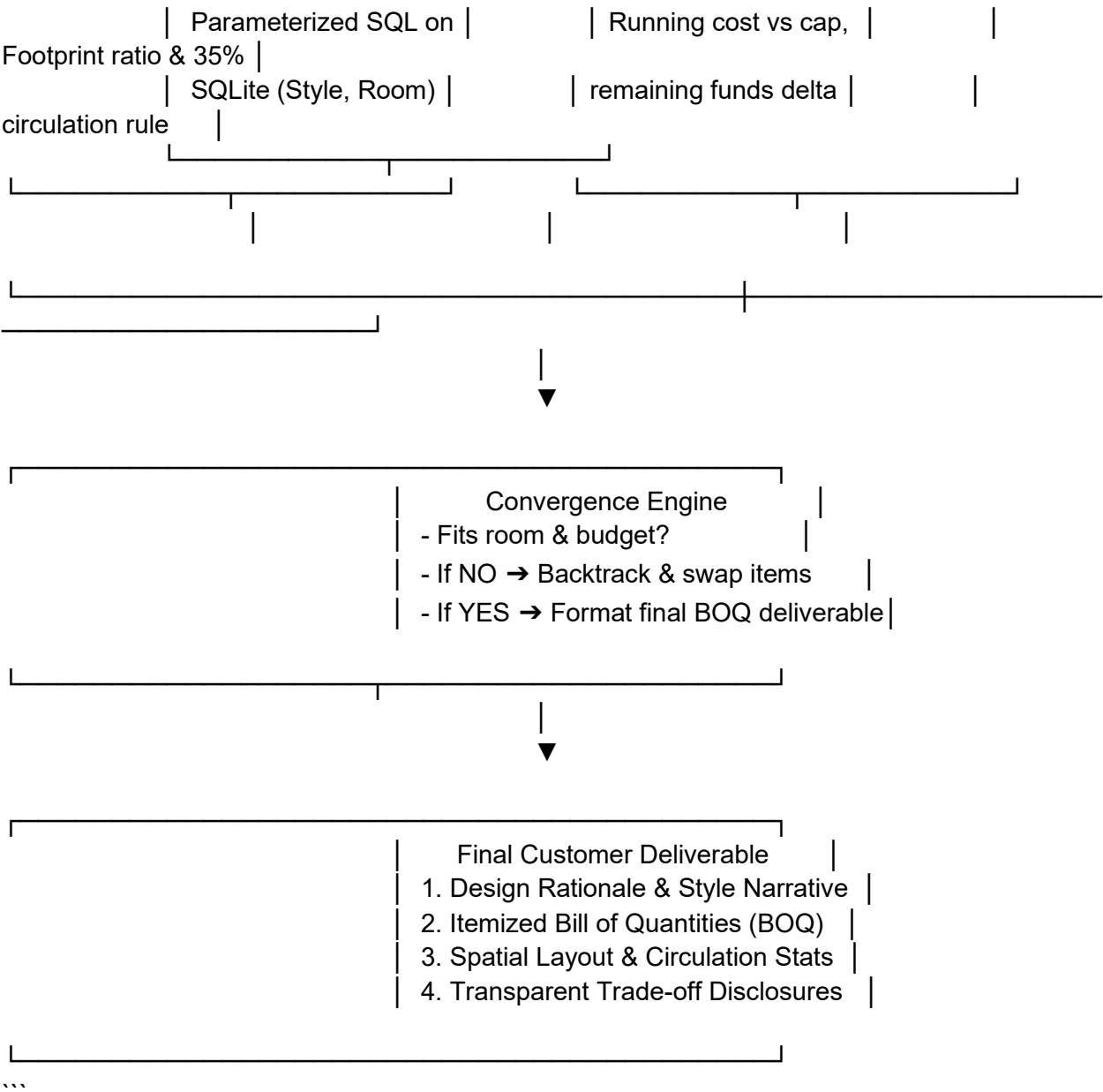

1. Executive Product Architecture Overview

The system is an autonomous, agentic reasoning pipeline designed for **Interior Company x Blocks**. It converts unstructured or semi-structured customer room briefs into budget-compliant, physically viable interior design plans and an itemized Bill of Quantities (BOQ).

Unlike single-shot chatbots that hallucinate products from memory, this agent operates within a **ReAct (Reasoning + Acting) loop** using three deterministic tools backed by a local SQLite database (`interior_company_catalog.db`).

...





2. Database Schema & Data Cleansing Protocol

The single source of truth is `interior\_company\_catalog.db`.

Table: `catalog` (72 Items)

- * `item\_id` (TEXT, PK): E.g., `SOF-001`, `CFT-002`, `TVU-001`.
- * `category` (TEXT): E.g., `Sofa`, `Coffee Table`, `TV Unit`, `Rug`, `Floor Lamp`, `Pendant Light`.
- * `name` (TEXT): Product marketing title.
- * `style\_tags` (TEXT): Comma-separated (e.g., `Scandinavian, Minimalist`, `Industrial`,

`Bohemian`).

- * `price\_inr` (INTEGER, Nullable): Item price in INR.
- * `width\_cm`, `depth\_cm`, `height\_cm` (INTEGER, Nullable): Physical dimensions in cm.
- * `color\_finish` (TEXT): Finish specification.
- * `in\_stock` (INTEGER): `1` = Available, `0` = Out of Stock.
- * `lead\_time\_days` (INTEGER): Delivery timeline.
- * `room\_types` (TEXT): Comma-separated applicability (e.g., `Living Room`, `Bedroom`).

Data Cleansing & Defensive Handling Rules:

1. ****Unpriced Products (`price\_inr` IS NULL):****
*** *Rule:*** Never treat `NULL` as ₹0. Unpriced items cannot be silently added to a committed BOQ. If selected, the agent must flag them with an estimated market allowance and label the row as `Price on Request / Awaiting Vendor Quote`.
2. ****Missing Dimensions (`width\_cm` or `depth\_cm` IS NULL):****
*** *Rule:*** When dimensions are `NULL`, the tool must impute conservative median footprints by category (e.g., Side Table: \$45 \times 45\text{ cm}\$; Floor Lamp: \$40 \times 40\text{ cm}\$; Decor/Plant: \$30 \times 30\text{ cm}\$) and mark the item for mandatory site verification.
3. ****Out-of-Stock Products (`in\_stock` = 0):****
*** *Rule:*** Filter out `in\_stock` = 0` by default. If lead time exceeds 30 days, do not include in rapid-move-in briefs unless the customer explicitly permits backorders.

3. Tool Specifications & Python Implementation Contracts

```
```python
import sqlite3
from typing import List, Dict, Any, Optional

DB_PATH = "interior_company_catalog.db"

def catalog_search(
 category: Optional[str] = None,
 style: Optional[str] = None,
 room_type: str = "Living Room",
 max_price: Optional[int] = None,
 in_stock_only: bool = True
) -> List[Dict[str, Any]]:
 """
 Query the SQLite catalog using parameterized filters.
 Enforces that the agent only accesses real, verified catalog items.
 """
 conn = sqlite3.connect(DB_PATH)
 conn.row_factory = sqlite3.Row
```

```

cursor = conn.cursor()

query = "SELECT * FROM catalog WHERE room_types LIKE ?"
params = [f"%{room_type}%"]

if category:
 query += " AND category = ?"
 params.append(category)
if style:
 query += " AND style_tags LIKE ?"
 params.append(f"%{style}%")
if max_price is not None:
 query += " AND (price_inr <= ? OR price_inr IS NULL)"
 params.append(max_price)
if in_stock_only:
 query += " AND in_stock = 1"

cursor.execute(query, params)
rows = [dict(r) for r in cursor.fetchall()]
conn.close()
return rows

```

```

def budget_calculator(
 selected_item_ids: List[str],
 total_budget_inr: int
) -> Dict[str, Any]:
 """
 Sums selected items, compares against total budget, and reports remaining balance.
 Detects unpriced items and prevents silent over-budget spending.
 """

 if not selected_item_ids:
 return {
 "total_spent": 0,
 "remaining_budget": total_budget_inr,
 "is_over_budget": False,
 "unpriced_items": [],
 "breakdown": []
 }

 conn = sqlite3.connect(DB_PATH)
 conn.row_factory = sqlite3.Row
 cursor = conn.cursor()

```

```

placeholders = ",".join(["?"] * len(selected_item_ids))
cursor.execute(f"SELECT item_id, name, category, price_inr FROM catalog WHERE item_id
IN ({placeholders})", selected_item_ids)
rows = cursor.fetchall()
conn.close()

```

```

total_spent = 0
unpriced = []
breakdown = []

```

```

for r in rows:
 price = r["price_inr"]
 if price is None:
 unpriced.append(r["item_id"])
 item_cost = 0
 else:
 item_cost = price
 total_spent += item_cost
 breakdown.append({
 "item_id": r["item_id"],
 "name": r["name"],
 "category": r["category"],
 "price": price
 })

```

```

remaining = total_budget_inr - total_spent
return {
 "total_budget": total_budget_inr,
 "total_spent": total_spent,
 "remaining_budget": remaining,
 "is_over_budget": remaining < 0,
 "overage_amount": abs(remaining) if remaining < 0 else 0,
 "unpriced_items": unpriced,
 "breakdown": breakdown
}

```

```

def layout_fit_check(
 room_length_cm: int,
 room_width_cm: int,
 selected_item_ids: List[str]
) -> Dict[str, Any]:
 """

```

Evaluates physical fit and spatial circulation.

Rule: Total furniture footprint must not exceed 35% of total room area.  
Ensures minimum 75cm - 90cm walkways.

"""

```
room_area_sqm = (room_length_cm * room_width_cm) / 10000.0
```

```
conn = sqlite3.connect(DB_PATH)
conn.row_factory = sqlite3.Row
cursor = conn.cursor()
placeholders = ",".join(["?"] * len(selected_item_ids))
cursor.execute(f"SELECT item_id, name, category, width_cm, depth_cm FROM catalog
WHERE item_id IN ({placeholders})", selected_item_ids)
items = cursor.fetchall()
conn.close()
```

```
total_footprint_sqm = 0.0
imputed_items = []
category_defaults = {
 "Sofa": (200, 90),
 "Coffee Table": (100, 60),
 "TV Unit": (160, 40),
 "Rug": (200, 140), # Rugs are floor coverings, discounted in clearance calculation
 "Armchair": (80, 80),
 "Floor Lamp": (40, 40),
 "Side Table": (45, 45),
 "Bookshelf": (80, 35)
}
```

```
for item in items:
```

```
 # Rugs do not block physical circulation height, footprint tracked separately
 if item["category"] == "Rug":
 continue
```

```
 w = item["width_cm"]
 d = item["depth_cm"]
 if w is None or d is None:
 default_w, default_d = category_defaults.get(item["category"], (50, 50))
 w = w or default_w
 d = d or default_d
 imputed_items.append(item["item_id"])
```

```
total_footprint_sqm += (w * d) / 10000.0
```

```
occupancy_ratio = total_footprint_sqm / room_area_sqm if room_area_sqm > 0 else 1.0
35% maximum furniture coverage threshold
```

```
fits_circulation = occupancy_ratio <= 0.35
```

```
return {
 "room_area_sqm": round(room_area_sqm, 2),
 "furniture_footprint_sqm": round(total_footprint_sqm, 2),
 "occupancy_ratio": round(occupancy_ratio, 3),
 "occupancy_percentage": f"{round(occupancy_ratio * 100, 1)}%",
 "fits_circulation": fits_circulation,
 "warning": None if fits_circulation else f"Furniture occupies {round(occupancy_ratio*100, 1)}% of floor area (Max allowable: 35%). Room will feel overcrowded and circulation corridors will be blocked.",
 "imputed_dimension_items": imputed_items
}
```

```

```

#### ### 4. BOQ Deliverable JSON Schema

Every successful agent execution returns this structured JSON:

```
```json
{
  "brief_id": "BR-01",
  "status": "SUCCESS",
  "design_concept": {
    "theme": "Calm Scandinavian Living Sanctuary",
    "palette_and_materials": "Light oak, oatmeal fabric, matte white, textured wool",
    "rationale": "Optimized for south-facing sunlight using light woods and reflective textures, maintaining calm, clutter-free circulation."
  },
  "boq": [
    {
      "item_id": "SOF-001",
      "category": "Sofa",
      "name": "Nordby 3-Seater Fabric Sofa",
      "dimensions": "210 x 88 x 82 cm",
      "finish": "Oatmeal grey",
      "price_inr": 62000,
      "in_stock": 1,
      "lead_time_days": 7
    }
  ],
  "financial_summary": {
    "budget_allocated_inr": 250000,

```

```
"total_spent_inr": 184000,
"remaining_budget_inr": 66000,
"budget_utilization_percentage": 73.6
},
"spatial_fit_summary": {
  "room_area_sqm": 17.28,
  "furniture_footprint_sqm": 4.12,
  "occupancy_percentage": "23.8%",
  "circulation_viable": true
},
"trade_offs_and_omissions": [
  "Prioritized high-quality 3-seater sofa and wool rug; excluded secondary accent armchair to
preserve open walking space."
]
}
...
```